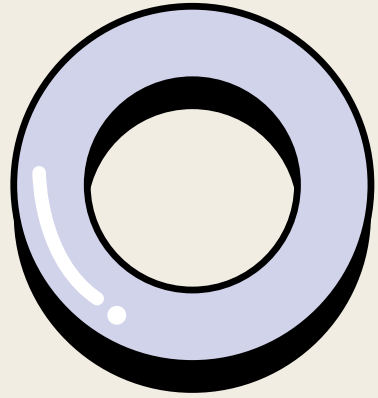




SOCIAL MEDIA ENGAGEMENT ANALYSIS USING PYTHON



**Presented by Riya Kiran and
Sanjitha G**

INTRODUCTION

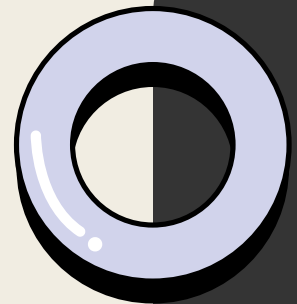
- **Social media is an important tool for marketing and brand communication.**
- **Businesses use platforms to connect with customers and promote products.**
- **Social media engagement includes likes, comments, shares, and views.**
- **Engagement metrics help measure content performance and audience interest.**
- **Analyzing engagement data helps improve marketing strategies and increase ROI.**
- **This project analyzes social media engagement using Python.**



Objectives of the project



- Analyze engagement patterns across platforms.
- Compare engagement across different content types.
 - Study likes, comments, shares, and views.



- Identify factors that drive higher engagement.
 - Present findings using data visualization.

Dataset Overview

Source: Kaggle

Total Rows: 10,000

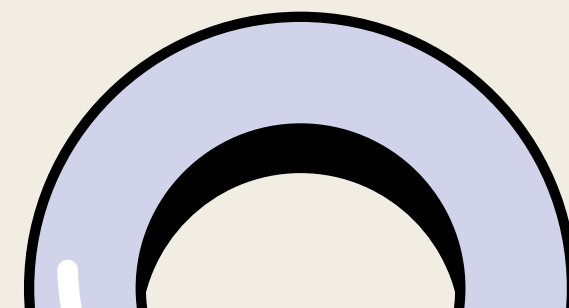
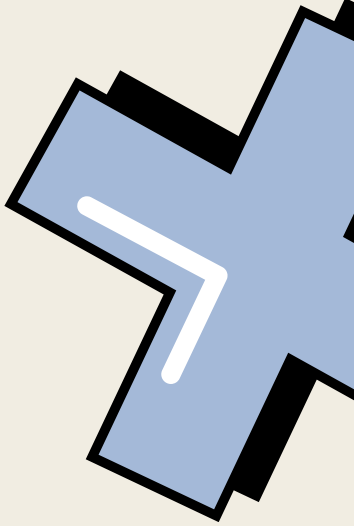
Data Type: Social media engagement data

Columns Used:

- Platform
- Post Type
- Likes
- Comments
- Shares
- Views
- Post length

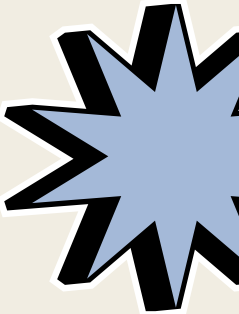
Tools and Technologies used

- Python – Used for data analysis.
- Pandas – Used for data cleaning and manipulation.
- Matplotlib – Used for basic data visualization.
- Seaborn – Used for advanced statistical charts.



Social media dataset

- Importing libraries and loading the dataset and checking it for null values



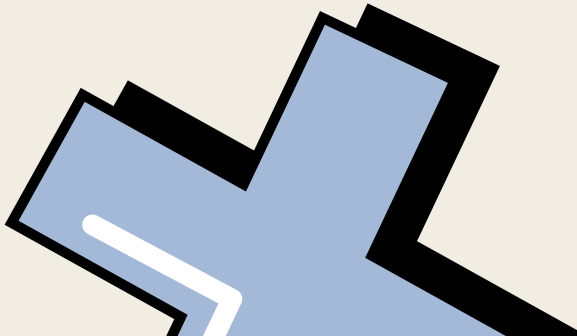
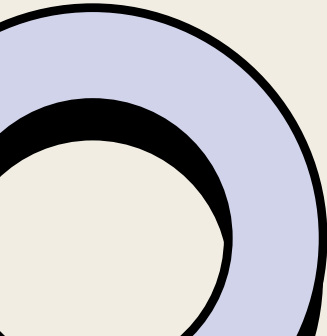
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sn
df=pd.read_csv("C:\\Users\\Dell\\Desktop\\ibm project\\social_media_engagement_dataset.csv")
df.head()
```

4.4s Open 'df' in Data Wrangler Python

	platform	post_type	# post_length	# views
0	Facebook	Text	62	91660
1	Instagram	Video	104	113115
2	Facebook	Video	46	36043
3	Facebook	Image	39	124886
4	Instagram	Video	42	82831

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   platform        10000 non-null  object
1   post_type       10000 non-null  object
2   post_length     10000 non-null  int64
3   views           10000 non-null  int64
4   likes           10000 non-null  int64
5   comments        10000 non-null  int64
6   shares          10000 non-null  int64
7   engagement_rate  10000 non-null  float64
dtypes: float64(1), int64(5), object(2)
memory usage: 625.1+ KB
```



Comparative Statistical Summary of Platforms

	post_length	views	likes	comments	shares	engagemen
count	3354	3354	3354	3354	3354	3354
mean	61.82	153182.84	8460.26	1058.92	727.66	0.07
std	33.45	85346.76	6579.49	953.41	692.86	0.03
min	5	5290	72	5	4	0.01
25%	32	79888.5	3038.25	343.25	216.25	0.04
50%	62	154173.5	3038.25	770.5	500	0.07
75%	91	226930.25	3038.25	1472.75	1018.75	0.09
max	119	299997	3038.25	5435	3861	0.13

INSTAGRAM

	post_length	views	likes	comments	shares	engagemen
count	3299	3299	3299	3299	3299	3299
mean	63.6	127698.51	7098.94	886.84	607.9	0.07
std	33.16	70813.93	5486.06	786.91	590.03	0.03
min	5	3132	49	6	4	0.01
25%	35	68717	2620.5	291	180	0.04
50%	64	127345	5835	651	413	0.07
75%	92	189553	10555.5	1249.5	865	0.09
max	119	249923	24843	4638	3464	0.13

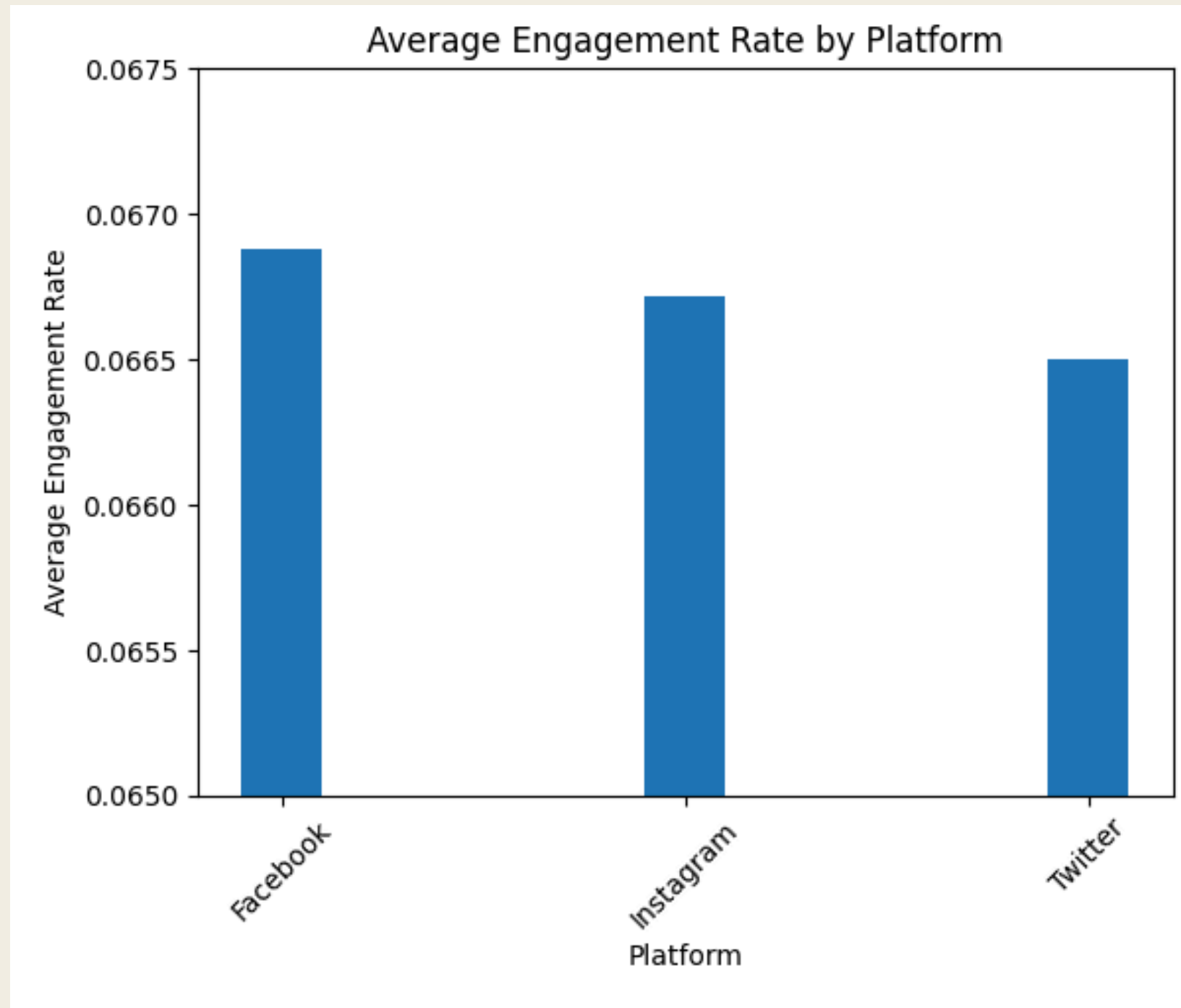
	post_length	views	likes	comments	shares	engagemen
count	3347	3347	3347	3347	3347	3347
mean	61.69	100527.9	5538.56	691.67	471.05	0.07
std	33.16	57053.81	4343.78	619.26	458.16	0.03
min	5	2059	35	3	2	0.01
25%	34	51672.5	1989	222	135	0.04
50%	62	101108	4518	505	321	0.07
75%	90	148952	8139.5	984.5	668.5	0.09
max	119	199999	19786	3675	2679	0.13

FACEBOOK

TWITTER

```
df_facebook=df[df["platform"]=="Facebook"]  
df_instagram=df[df["platform"]=="Instagram"]  
df_twitter=df[df["platform"]=="Twitter"]  
df_facebook.describe()  
df_instagram.describe()  
df_twitter.describe()
```

Average Engagement Rate by Platform



- Facebook demonstrates the highest average engagement rate, indicating stronger audience interaction per post.
- Instagram ranks second, showing competitive but slightly lower engagement compared to Facebook.
- Twitter records the lowest average engagement rate, suggesting relatively weaker overall interaction levels.
- Engagement differences across platforms are moderate, indicating relatively balanced performance overall.


```
platform_avg = df.groupby('platform')['engagement_rate'].mean()

highest_value = platform_avg.max()
highest_platform = platform_avg.idxmax()

print("Highest platform:", highest_platform)
print("Engagement rate:", highest_value)

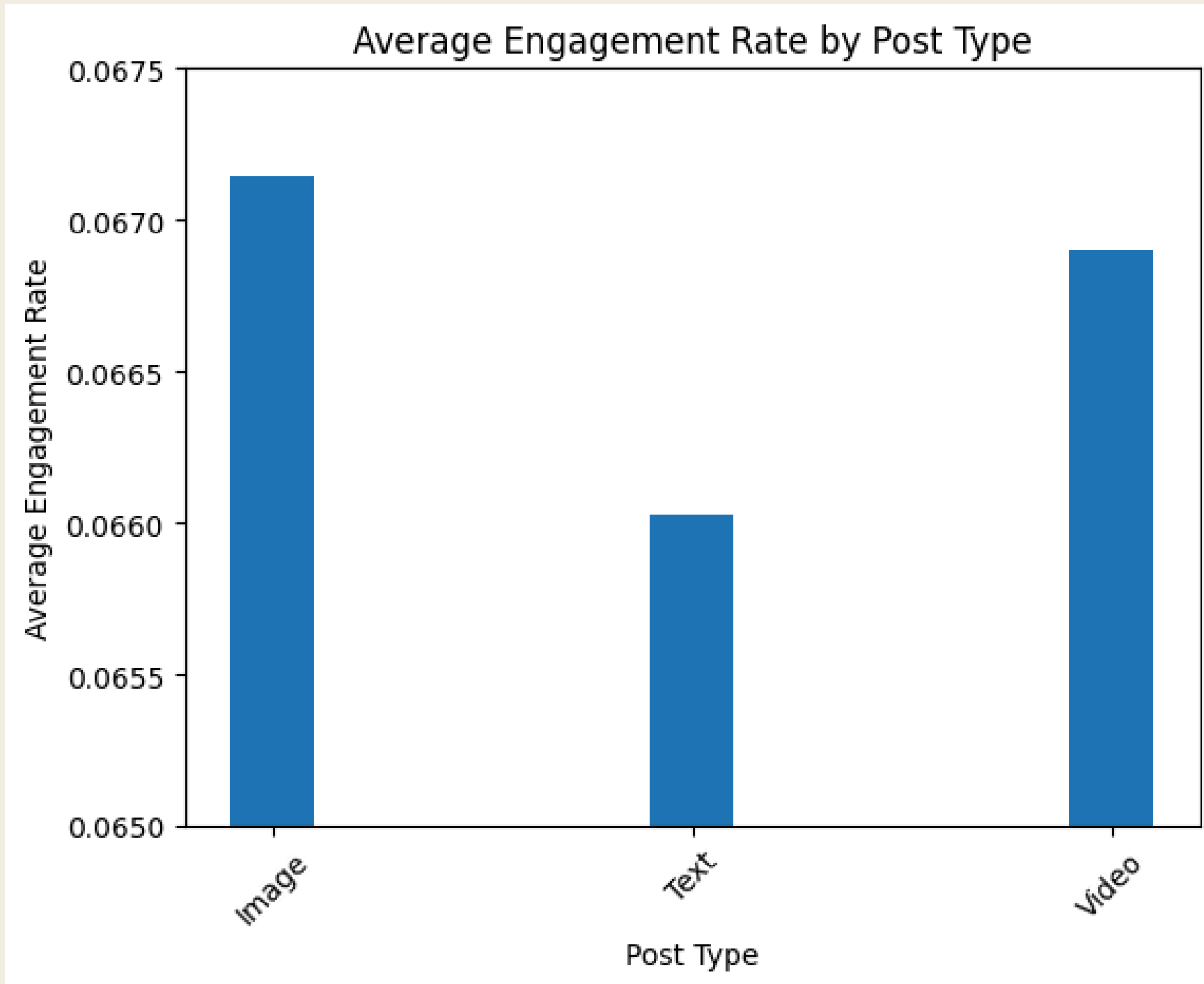
lowest_value=platform_avg.min()
lowest_platform=platform_avg.idxmin()
print("lowest value",lowest_value)
print("engagement rate",lowest_platform)
```

```
import matplotlib.pyplot as plt

platform_avg = df.groupby('platform')['engagement_rate'].mean()

plt.figure()
plt.bar(platform_avg.index, platform_avg.values,width=0.2)
plt.ylim(0.0650,0.0675)
plt.title("Average Engagement Rate by Platform")
plt.xlabel("Platform")
plt.ylabel("Average Engagement Rate")
plt.xticks(rotation=45)
plt.show()
```

Average Engagement Rate by Post type



- Image-based posts generate the highest average engagement, indicating strong audience interaction with visual content.
- Video content ranks second, showing effective but slightly lower engagement compared to images.
- Text-only posts record the lowest engagement rate, suggesting limited interaction relative to visual formats.
- The results highlight the importance of visual content in driving audience engagement.

```
post_avg = df.groupby('post_type')['engagement_rate'].mean()
print(post_avg)

# Highest
highest_value = post_avg.max()
highest_content = post_avg.idxmax()

print("Highest content type:", highest_content)
print("Engagement rate:", highest_value)

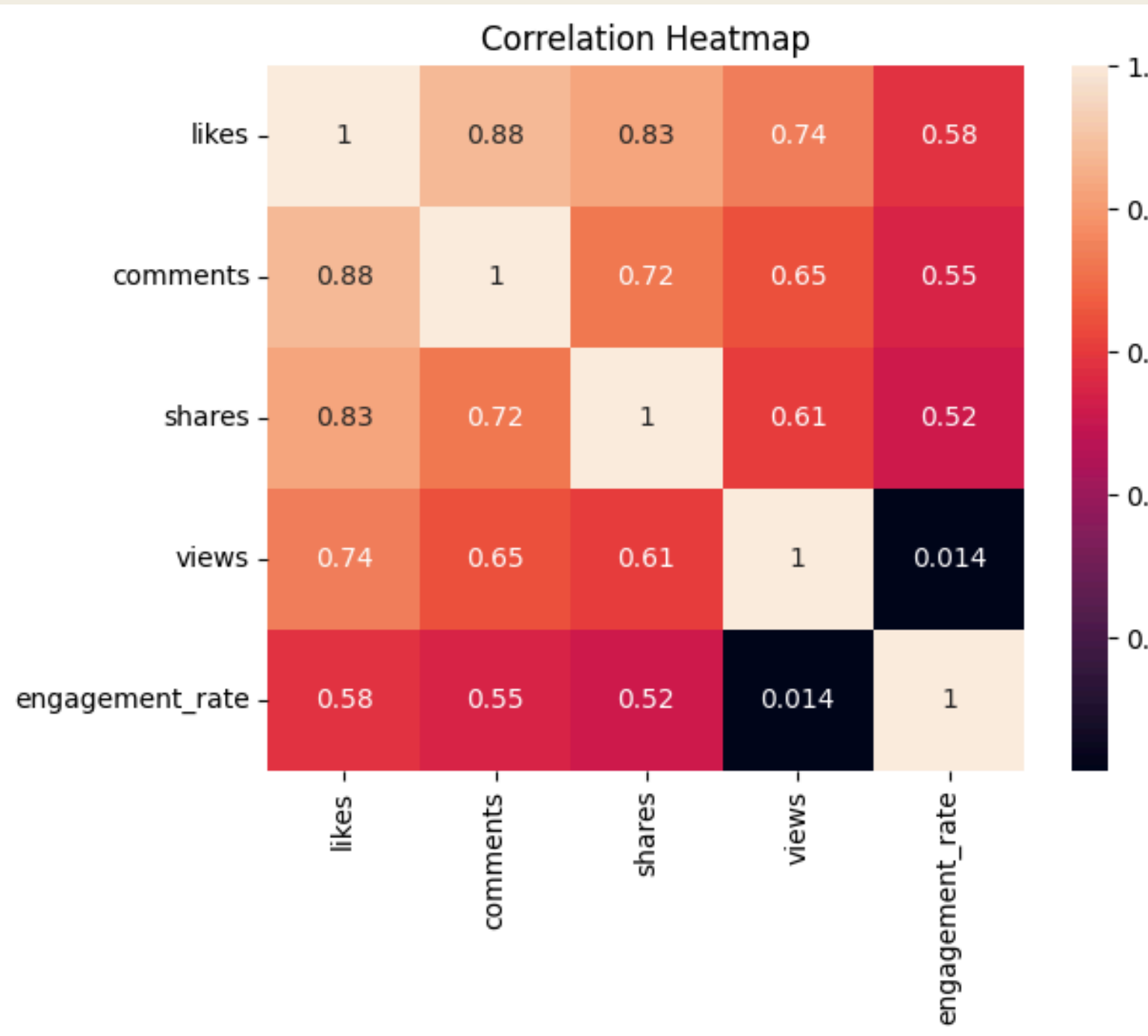
# Lowest
lowest_value = post_avg.min()
lowest_content = post_avg.idxmin()

print("Lowest content type:", lowest_content)
print("Engagement rate:", lowest_value)
```

```
import matplotlib.pyplot as plt

plt.figure()
plt.bar(post_avg.index, post_avg.values,width=0.2)
plt.ylim(0.0650,0.0675)
plt.title("Average Engagement Rate by Post Type")
plt.xlabel("Post Type")
plt.ylabel("Average Engagement Rate")
plt.xticks(rotation=45)
plt.show()
```

Correlation Analysis of Engagement Metrics



- Likes & Comments Have a Very Strong Relationship (0.88)
- Likes & Shares Are Strongly Correlated (0.83)
- Views Have Moderate Correlation With Likes, Comments & Shares (0.61–0.74)
- Engagement Rate Has Weak to Moderate Correlation With Likes, Comments & Shares (0.52–0.58)
- Very Important Finding: Views & Engagement Rate Have Almost No Correlation (0.014)

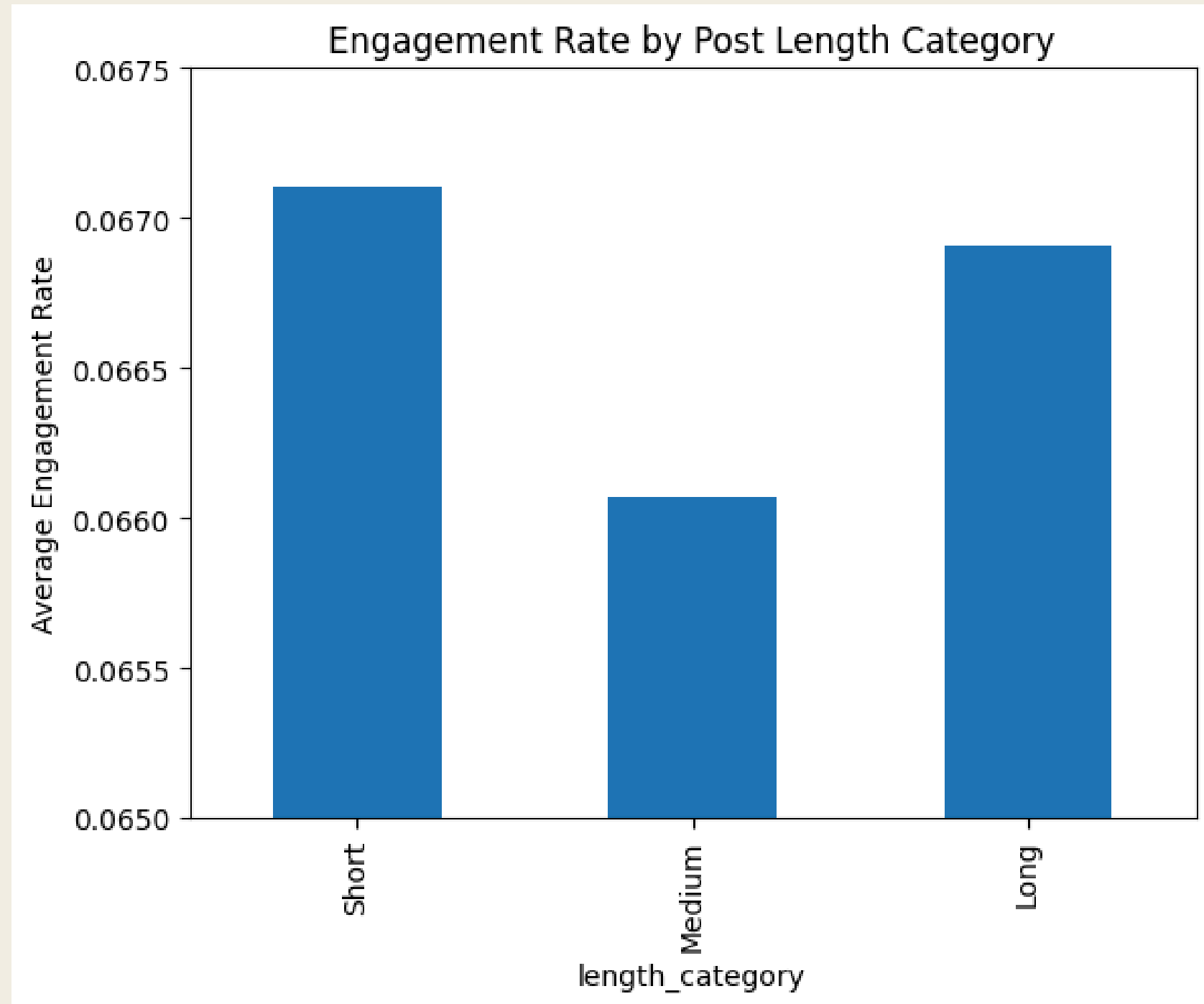
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure()
corr = df[['likes', 'comments', 'shares', 'views', 'engagement_rate']].corr()

sns.heatmap(corr, annot=True)

plt.title("Correlation Heatmap")
plt.show()
```


Average Engagement Rate by Post length

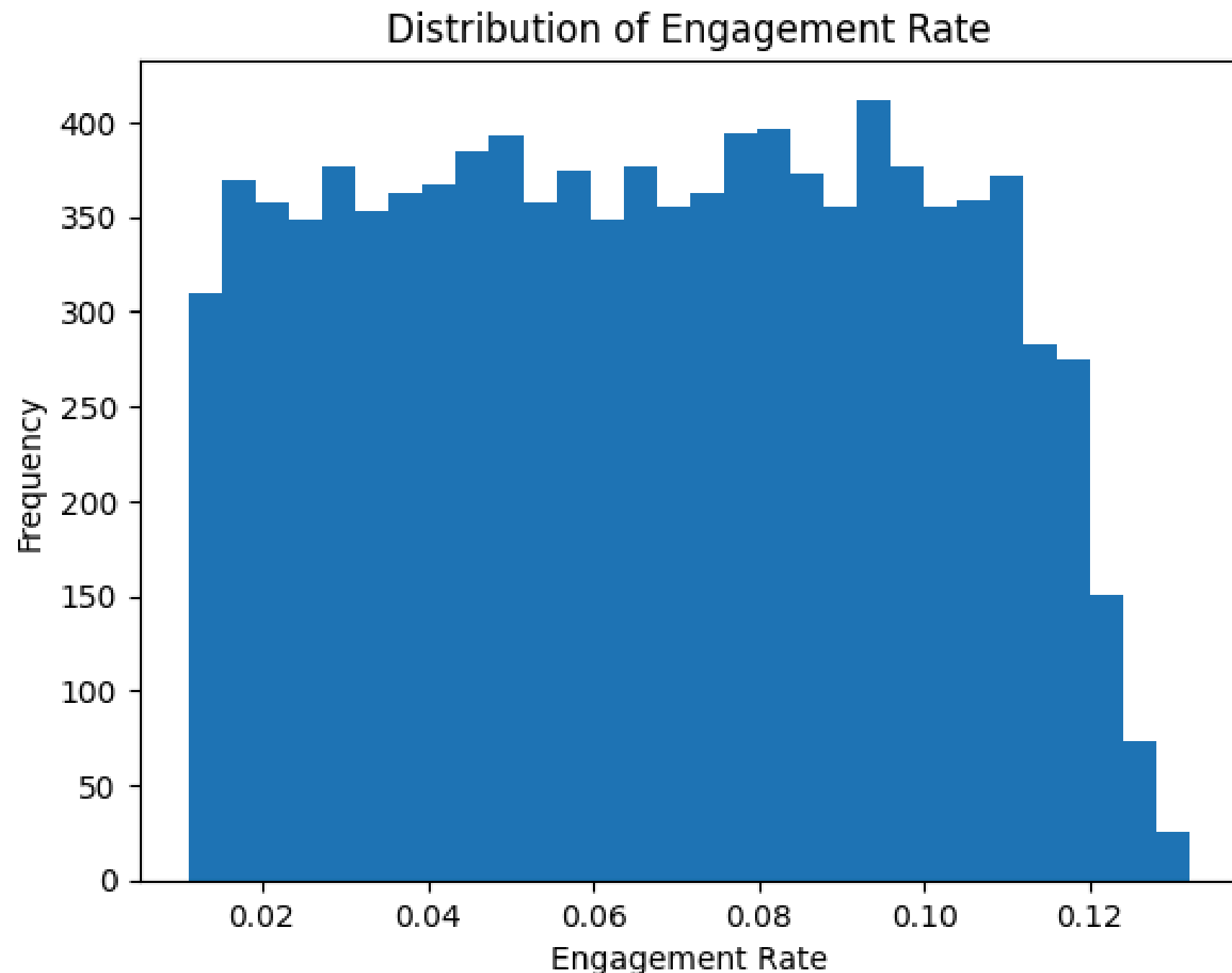


- Short-length posts generate the highest average engagement rate
- Long posts perform better than medium-length posts
- Medium-length posts record the lowest engagement
- The results suggest that content effectiveness may follow a non-linear pattern rather than a simple “longer is better” trend.

CODE

```
df['length_category'] = pd.cut(df['post_length'], bins=3,  
                                labels=['Short', 'Medium', 'Long'])  
  
df.groupby('length_category')['engagement_rate'].mean().plot(kind='bar')  
plt.ylim(0.0650, 0.0675)  
  
plt.title("Engagement Rate by Post Length Category")  
plt.ylabel("Average Engagement Rate")  
plt.show()
```

Distribution of Engagement Rate



- Most posts fall between 3% and 11% engagement.
- The highest concentration is around 7%–10%.
- Very low engagement ($<2\%$) is uncommon.
- Very high engagement ($>12\%$) is also uncommon.
- The distribution looks fairly even — no extreme spikes.


```
plt.figure()  
plt.hist(df['engagement_rate'], bins=30)  
plt.title("Distribution of Engagement Rate")  
plt.xlabel("Engagement Rate")  
plt.ylabel("Frequency")  
plt.show()
```

Key Insights

- Short-form video content (Reels) consistently achieved the highest engagement rates compared to static posts.
- Instagram demonstrated the most consistent engagement performance with a stable median and lower variability.
- Video-based content generated higher shares, suggesting stronger audience interest and content relatability.
- A small number of high-performing posts significantly influenced overall engagement metrics, indicating the presence of viral spikes.

Business Recommendations

- Prioritize short-form video content to maximize engagement and audience interaction.
- Schedule posts during high-activity periods, particularly weekends, to increase visibility and reach.
- Optimize captions and call-to-actions to encourage comments and discussions.
- Incorporate interactive elements such as polls, Q&A sessions, and user-generated content.
- Analyze high-performing posts and replicate successful content patterns to improve consistency.

Conclusion

- Social media engagement is influenced by content format, platform choice, and posting timing.
- Short-form video content drives stronger audience interaction and shareability.
- Consistent engagement patterns are crucial for sustainable growth.
- Data-driven strategies enable better optimization of digital marketing efforts.